ENGINEERING LEARNING ACADEMY

FR-018 M3 — Production Hardening & Source Expansion

FR018 — sources/optional/m3.md

Masterclass source package snapshot

Functional Requirement	FR-018
Edition	Engineering Learning Academy — Source Package snapshot
Status	**Complete (M3)** — **GO WITH REVISED SCOPE
Audience	Academy generation and study (snapshot; SoT remains authoritative)
Document type	Source package snapshot (PDF)

Contents

FR-018 M3 — Production Hardening & Source Expansion

Status: Complete (M3) — **GO WITH REVISED SCOPE**
Date: 2026-08-07
Phase: Horizon 1B (lead FR)
Preceding: M0, M1, M2, post-M2 live validation
ADR: ADR-010

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1. Executive summary

M3 production-hardens the M2 URL path without scrape theatre.

Area	Outcome
SEEK	**Production ready** — live `UrllibHttpClient` acquire succeeds (title + body + provenance)
SSL	**Resolved in code** — `truststore.SSLContext` (OS trust store); root cause was conda/ `SSL_CERT_FILE` CA path vs OpenSSL 3
LinkedIn	**Evidence-based unsupported for reliable URL acquire** — classify OK; live redirects to listings; fail closed
Indeed	**Evidence-based unsupported for reliable URL acquire** — Cloudflare `HTTP 403`; no bypass
New sources	None added (no clean lawful HTTP board beyond SEEK)
Ingress	Remains thin (ADR-010 unchanged)
Horizon 1A	Unchanged

Verdict: GO WITH REVISED SCOPE — ship SEEK as the production URL source; keep LinkedIn/Indeed as attempt/fail-closed; next volume path is email-alert acquisition (M4+), not Playwright.

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2. SEEK production readiness

Concern	Change / evidence
Canonical URL	Host variants (`www.seek.com.au` , `au.seek.com`) → stable `https://www.seek.com.au/job/<id>`
Identity	`platform_job_id` + canonical share across hosts; definite skip still FR-009 reuse
Provenance	`source_kind=url` , canonical `source_url` / `source_identifier` , `acquired_at`
Extract	Title cleaning (`… Job in … - SEEK` → role title); company left unset when only board brand is available (fail closed vs inventing employer)
Redirect	`www` → `au.seek.com` recorded as warning; acquisition still succeeds
Errors / UX	Clearer CLI hints on failure; offline fixtures unchanged
Live	`UrlAcquisitionAdapter` + `UrllibHttpClient` acquired SEEK job `93312273` (title `AI Engineer` , raw ~1.7k chars)

Owner experience: `cic opportunity discover <seek-job-url>` is the production paste path once analysis credentials are present (or `--offline-fixtures` for smoke).

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3. SSL investigation

Root cause (environment, not architecture)

Fact	Detail
Symptom	`CERTIFICATE_VERIFY_FAILED: Basic Constraints of CA cert not marked critical`
Default verify paths	Conda `Library/ssl/cacert.pem` ; `SSL_CERT_FILE` pointed at that bundle
`certifi` alone	Still failed (`unable to get local issuer certificate`)
`truststore.inject_into_ssl()`	Still failed (global inject does not override broken env CA for all hosts)
`truststore.SSLContext(PROTOCOL_TLS_CLIENT)`	**Succeeds** — uses Windows OS certificate store
`curl.exe`	Already succeeded (same OS store)

Conclusion: Local Python was verifying against a conda CA bundle that OpenSSL 3 rejects for these hosts. System TLS was fine. This is an **environment/configuration defect**, not a board or adapter design defect.

Solution

- Production code: `UrllibHttpClient` uses `build_default_ssl_context()` → prefer `truststore.SSLContext`, else stdlib default. Verification remains **on**.
- Dependency: declare `truststore>=0.10.0` in `pyproject.toml` (already present in the owner env; now explicit).
- No certificate pinning, no `CERT_NONE`, no anti-bot bypass.

Production code changes required?

Yes, narrowly: discovery HTTP client must construct an OS-backed SSL context. No Horizon 1A changes. CLI `inject_into_ssl()` for OpenAI remains separate and optional.

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4. LinkedIn investigation

Approach	Result
Plain HTTP GET of `/jobs/view/<id>`	Often **200** but final URL `?trk=expired_jd_redirect` → search listing
Slug URLs `/jobs/view/…-<id>`	**Normalised** (M3) to numeric canonical — classify works; acquire still fails closed on listing
Metadata / public endpoints	No stable undocumented public job JSON used — would be brittle / ToS-risk
Login / session	Rejected — out of principles
Playwright	Rejected — automation theatre

Recommendation: Keep LinkedIn on the allow-list as an **attempt** path with fail-closed quality gates (`expired_jd_redirect`, listing titles, non-`/jobs/view/` finals). Do **not** claim production LinkedIn URL acquire. Prefer **email-alert / paste** later for LinkedIn volume.

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5. Indeed investigation

Approach	Result
Plain HTTP `viewjob?jk=`	**HTTP 403** Cloudflare challenge
Alternate headers / UA	Not pursued as bypass theatre
Public API	No owner-approved Indeed API integration in scope
Playwright / challenge solve	Rejected

Recommendation: Keep Indeed classify + fail-closed on 403/challenge. Document limitation. Email alerts / paste for Indeed volume later. **Do not bypass anti-bot.**

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6. Newly supported sources

None. No additional board met "robust, lawful, maintainable plain HTTP" without browser automation. Architecture remains extensible via `SUPPORTED_PLATFORMS` + `derive_source_facets` + extractor regions.

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7. Sources remaining unsupported (URL path)

Source	Classification	Why
LinkedIn (reliable live job body)	Attempt / fail closed	Listing redirects & walls without login/browser
Indeed (reliable live job body)	Attempt / fail closed	Cloudflare challenge on plain GET
Company careers pages	`unsupported_source`	Not allow-listed; no generic site scraper

Paste/export (Horizon 1A) remains the universal fallback.

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8. Engineering decisions

Decision	WHY	Alternative	Rejected because	Principle	Interview takeaway
`truststore.SSLContext` in fetch client	Fix env TLS without disabling verify	Ignore SSL / CERT_NONE	Security theatre & false “board blocked” signal	Fail closed; validate first	Separate env defects from product defects
Stable SEEK `www.seek.com.au` canonical	Idempotency across host redirects	Preserve raw netloc	Duplicate false misses across www/au	Explicit provenance	Canonicalise identity, not page chrome
LinkedIn listing/redirect gate	Prevent false Opportunities from search HTML	Accept any long HTML	Pollutes SoT	Fail closed	HTTP 200 ≠ usable job ad
No Playwright for LinkedIn/Indeed	Lawful maintainability	Browser automation	Scrape theatre; fragile	No automation theatre	Unsupported is a valid engineering answer
No new boards	Dual-value / evidence	Add Jora/etc. without live proof	Scope expansion without validation	Validate first	Extensibility ≠ premature allow-list growth
Ingress stays thin	ADR-010	Fat discovery orchestrator	Reopens FR-008	Thin interfaces; reuse	Coordinator ≠ second OS

9. Test results

Suite	Result
`tests/unit/discovery/`	**47 passed**
`tests/unit/opportunities/test_identity.py`	passed (incl. SEEK canonical + LinkedIn slug)
Broader: discovery + identity + duplicates + orchestration models + FR-008 functional	**105 passed**

New coverage: SSL context construction; SEEK host canonical; LinkedIn slug classify; listing/redirect fail-closed; Indeed 403 mapping; ingress acquire + definite skip; injected SSL context on `urlopen`.

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10. Manual validation (M3 live)

Client: production `UrllibHttpClient` (truststore SSL). No Playwright. No Opportunity persist required for this table (acquire-only).

Source	Classify	Fetch / extract	Support
SEEK `…/job/93312273`	seek / id / canonical	**OK** — title `AI Engineer`, usable body	**SUPPORTED**
LinkedIn `/jobs/view/4429615445`	linkedin	`blocked_response` (expired redirect listing)	**UNSUPPORTED (reliable)**
LinkedIn slug URL	linkedin (slug→id)	`blocked_response`	**UNSUPPORTED (reliable)**
Indeed `jk=6449f2b22e094d45`	indeed	`http_error` HTTP 403	**UNSUPPORTED (reliable)**
Thoughtworks careers	unsupported_source	n/a	**UNSUPPORTED**

Owner UX: failure messages include paste/export hint; unsupported hosts explain allow-list.

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11. Documentation updated

- This report; changelog § 1.120
- ADR-010 M3 note; testing strategy; implementation notes
- Functional specification (FR-018 status); roadmap; phase history
- README / AGENTS / repository guide (status only)

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12. Learning outcomes

1. **Environment vs platform:** Conda CA + OpenSSL 3 looked like “boards block Python”; OS trust store proved otherwise for SEEK.
2. **200 is not success:** LinkedIn listing redirects are the classic false-positive acquisition hazard.
3. **Unsupported is deliverable:** Documented LinkedIn/Indeed limits beat brittle bypasses.
4. **Thin ingress held:** Hardening stayed in HTTP/classify/extract — not a second orchestrator.

13. Recommendation

M3 COMPLETE — GO WITH REVISED SCOPE

- Treat **SEEK URL discover** as production-ready.
- Leave LinkedIn/Indeed as classify-and-fail-closed attempts.
- **M4 direction:** email-alert (or export) acquisition for volume — not Playwright, not LinkedIn/Indeed scrape escalation.
- Stop before M4.

14. Final repository status

Item	Status
Horizon 1A	Frozen / unchanged
FR-018 M0–M2	Accepted
FR-018 M3	**Complete — GO WITH REVISED SCOPE**
M4	Not started

M3 COMPLETE — GO WITH REVISED SCOPE